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Vision Transformers for the Self-Supervised Classification of Images of Rock Samples

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Abstract

The subsurface disciplines are always in a quest for analogs, whether it be geologic, petrophysical or engineering related. The classification of rock sample images provides a valuable start point in the search for analogs by providing an understanding of microstructural similarities and differences between different plays. Convolutional Neural Networks (CNN) are the dominant approach for these applications and are well-suited for these tasks given the inductive biases inherent to the process of training, which is invariance of the images under translation or rotation. Processing these images, however, can take an inordinate amount of time because of the need for proper annotation, which is extremely time consuming and resource intensive with conventional supervised learning approaches. Self-supervised learning, remarkably, is capable of distilling the information content of images to find digital analogs rapidly.

Our dataset comprises 22,440 FIB-SEM images collected from 22 shale plays across North and South America. We explore a vision transformer-based architecture for the classification of these SEM images. Our goal is to identify the source of the image from among the 22 different unconventional plays. From a training set, we successively feed unlabeled images to independent teacher and student vision transformer (ViT) networks that learn from each other and progressively get better at identifying similarities and distinctions in the microstructure. We then exploit this by pairing the ViT-based algorithm with a k-Nearest Neighbors (k-NN) classifier that groups the images.

The surprising result was the fidelity with which the features associated with any given play are automatically distilled from the images over the training period. This enabled a very high accuracy and low misclassification rate among 22 different plays. Without prior training on labeled data, the model successfully identified key image features that guide its classification through attention maps. Interestingly, there were a few misclassified samples from pairs of plays that share common features and form digital analogs.

This work constitutes the first approach for self-supervised classification of images of rock samples. The self-supervision automatically extracts meaningful information that we discuss in the paper in the form of attention maps. These maps contain microstructural details that enable rapid and efficient classification, but more importantly, provide a pathway for interpretability for the petrologist or geologist concerned.

Introduction

Focused Ion Beam–Scanning Electron Microscopy (FIB-SEM) is an essential tool in Digital Rock Physics (DRP), enabling high-resolution computational analysis of rock properties (Andrä et al., 2013). Widely used for visualizing, modeling, and characterizing nanoscale features in shales (Curtis, 2010; Lemmens et al., 2011; Shabro et al., 2014; Goral et al., 2015; Cudjoe et al., 2019; Cudjoe et al., 2020), FIB-SEM also supports a broad range of petrophysical analyses. These include pore-throat diameter distributions, pore connectivity, and tortuosity (Dewers et al., 2012), quantitative evaluations of porosity, permeability, and transport properties (Yoon & Dewers, 2013), qualitative assessments of pore morphology (Ma et al., 2015), detailed pore classification and comparative analyses of shale formations for gas storage potential (Zhou et al., 2016) and permeability prediction (Avdonin et al., 2021).

Image classification is a core task in computer vision, typically involving the categorization of images based on textures and features (Feng et al., 2019; Fang et al., 2020). In the context of FIB-SEM, this involves labeling images of shales with the geological formation they were sourced from. Traditionally, this task has relied on manual interpretation that is both time-intensive and prone to subjective bias. In contrast, deep learning models enable automated classification that is faster, more consistent, and more scalable. Correct classifications capture distinctive microstructural features across different shale plays, while misclassification can reveal microstructural similarities, possibly indicating shared petrophysical characteristics (H. Zhang et al., 2021). Systematic classification also facilitates comparative analysis across plays, (Keller et al., 2011). Classifying FIB-SEM images by geological play also enables the use of tailored models to analyze pore structures, organic matter distributions, and mineral compositions, which are the critical inputs for petrophysical modeling and reservoir evaluation (Loucks et al., 2009; Passey et al., 2010; Chalmers et al., 2012).

Classifying FIB-SEM images of shale samples presents a distinct set of challenges. The imaging process itself varies from image to image, with the use of different devices, different imaging parameters and possibly different operators. Machine learning models, unfortunately, can potentially learn from imaging artifacts thereby compromising the classification process. Standardization of the images is therefore necessary. Secondly, the microstructural similarity between different formations can further complicate classification tasks (Knaup et al., 2019; Knaup et al., 2022). Fig. 1 shows FIB-SEM images from the Montney, Haynesville, and Eagle Ford formations. Because the Montney and Haynesville samples appear visually similar, accurate classification can be more challenging.

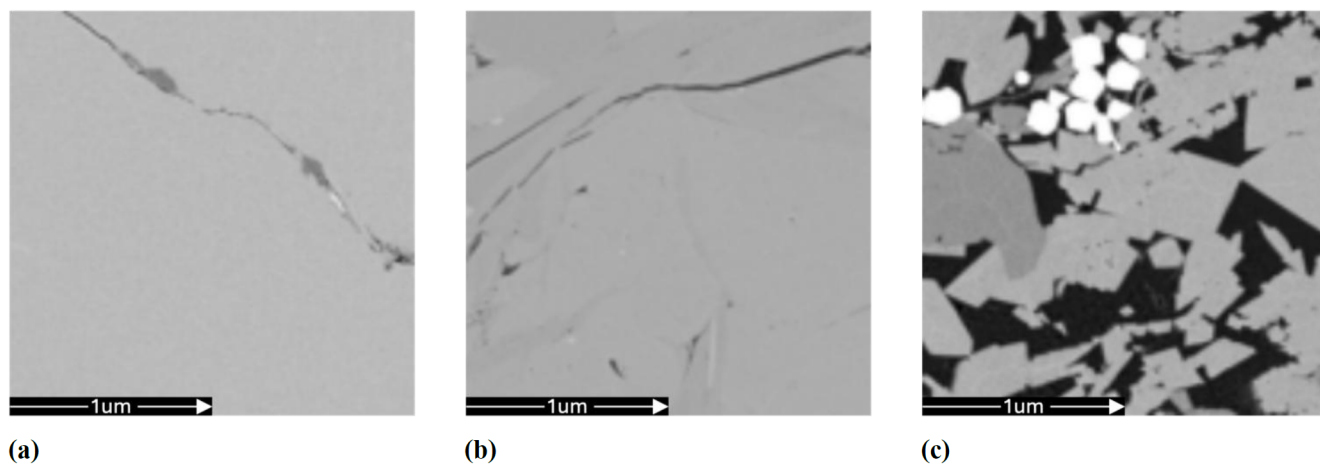


Figure 1—This figure presents examples of SEM images from (a) Montney, (b) Haynesville, and (c) Eagle Ford Gas plays. Misclassification between Montney and Haynesville often occurs due to the absence of distinct features in both formations.

The second challenge lies in the limitations of existing classification methods. Convolutional Neural Networks (CNN) have been the dominant architecture for image classification (Krizhevsky et al., 2012; Hsu et al., 2020; Rakesh et al., 2022) because they are translation-invariant and harness the differences and similarities between spatially adjacent pixels. CNNs have been extensively applied to geological image classification tasks, including shale FIB-SEM images (Knaup et al., 2019; H. Zhang et al., 2021; Knaup et al., 2022), core samples (Wang et al., 2014; Kathrada and Adillah, 2019; B. Zhang et al., 2021; Carpenter, 2024), and thin sections (Mosser et al., 2022; Tolstaya et al., 2023).

However, CNNs have limited capacity to capture long-range spatial dependencies (Raghu et al., 2021; Azad et al., 2024). In contrast, Vision Transformers (ViT) have emerged as a more promising tool, capturing long-range relationships through the concept of self-attention (Dosovitskiy et al., 2020; Caron et al., 2021). Nonetheless, both CNN and ViT classifiers require a robust training process that depends on the availability of large, high-quality labeled datasets. Unfortunately, this is an especially demanding requirement in domains like FIB-SEM shale imagery, where expert labeling is both time-consuming and resource-intensive.

This study overcomes the need for large, manually labeled datasets by using a self-supervised learning framework that relies on the use of vision transformers (ViTs). While the concept of self-supervised learning has also been applied with CNNs, they have seen relatively little success. Kong et al. (2023) report accuracies as low as 32% on the well-known MNIST dataset for handwritten digit classification. This is likely because CNNs are adept at capturing local or short-range relationships but weaker at quantifying long-range spatial dependencies. In a previous paper (Mohammad et al 2025), we document self-supervised learning for segmentation tasks, and show how self-supervision using vision transformer architecture outperforms CNN-based segmentation (Knaup et al. 2022).

Description of the Dataset

The dataset used in this study comprises 22,440 FIB-SEM grayscale BSE images collected from 22 shale plays across North and South America and Europe. Each image was standardized to 8-bit depth, resized to 224×224 pixels and rescaled to a resolution of 50 nm per pixel. The dataset was split into 80% for training and 20% for testing and prediction. To further enhance consistency, all images were normalized based on their mean and standard deviation.

Description of a Vision Transformer

We start by introducing the architecture and core principles of a vision transformer (ViT) which forms the backbone of the self-supervised classifier used in this paper.

Originally introduced for Natural Language Processing (NLP) tasks such as machine translation, Transformers (Vaswani et al., 2017) laid the groundwork for advanced language models like GPT (Radford, 2018) and BERT (Kenton and Toutanova, 2019) because they excel at processing sequential data, such as words in a sentence. This architecture was later adapted for computer vision tasks through Vision Transformers (ViT) (Dosovitskiy et al., 2020), which treat image patches as sequential tokens. The self-attention mechanism (Vaswani et al., 2017) in ViT then enables quantifying relationships between adjacent image patches as well as those farther away, capturing both local and global information, and providing a more complete spatial representation of the image.

The original ViT architecture, as illustrated in Fig. 2 (Dosovitskiy et al., 2020), serves as the foundation for our classification algorithm. In the following section, we provide a brief overview of the image processing steps involved in enabling classification.

1. **Generating Input Patches:** ViT begins by dividing an input image into small, non-overlapping patches. As illustrated in Fig. 2, the image is split into 9 distinct patches, each treated as an individual token. Each patch is then flattened from its original 2D pixel format into a 1D vector. This vector is

subsequently passed through a linear projection layer, which applies trainable weights and biases for dimensionality reduction, facilitating efficient processing in subsequent layers.

2. **Encoding Positional Information:** The linearly projected tokens lack inherent information about their spatial positions within the image. To preserve the spatial structure, a positional encoding vector is added to each token. These enriched vectors, now containing both content and positional context, are then passed to the Transformer encoder as input.
3. **The Classification Token:** A special classification token, denoted as [CLS], is prepended to the sequence of image patches, assigned as Patch 0 and bringing the total number of tokens to 10. Initially initialized with random values, the [CLS] token learns to aggregate information from all other tokens as it passes through the trained ViT layers. The final representation of the [CLS] token is then used for downstream classification tasks.
4. **The Attention Mechanism:** Attention can broadly be defined as the relationship between pixels or patches within an image. It was originally defined to extract relationships between words in a sentence (Vaswani et al., 2017) and was later adapted for images (Vaswani et al., 2017; Dosovitskiy et al., 2020). The computation of attention is as follows: The attention head contains three weight matrices called the Query (Q), Key (K), and Value (V) matrices. The matrices represent networks with learnable weights. Each matrix operates upon the input tokens (with the positional encoding) to generate the Query, Key and Value vectors. For each input token, the Q matrix operating on the input token generates a Query vector. For all the N input tokens, we also get N Key vectors. To assess the similarity between the 4th token and all other tokens, we ‘compare’ the 4th Query vector with all the other N Key vectors. This is done by using a dot product between the Query and Key vectors. This scalar similarity index is multiplied with each Value (V) token and the result added together to get the attention map corresponding to 4th input token. In other words, a weighted average of the Value vectors is executed where the weights are the dot product between the Query and Key vectors. This forms the attention matrix.
 - a. Conceptually, the process can be described with an example: At the start of training, the weight matrices for Query, Key, and Value are initialized randomly, and their roles are arbitrary.
 - b. As training progresses, the Query matrix perhaps learns to identify pores and its spatial juxtaposition to the organic and inorganic material, adjusting its weights accordingly. The Key matrix also learns to represent relevant features such as pores and their spatial arrangement in the image.
 - c. For an image from a specific play with a specific spatial arrangement of pores, organic and inorganic material, the Query vector with the highest similarity to a specific Key vector passes on information that is relevant for downstream processing.
 - d. The Value matrix is trained to preserve the original information of each token. Once a high Query-Key similarity is found, the corresponding Value vector is weighted accordingly, ensuring that relevant information is retained in the final attention output.
5. **Multi-Head Attention:** Rather than relying on a single attention mechanism, ViTs typically use multi-head attention. With two attention heads, the second head effectively compute the attention of the first attention head, which in turn captures a wider scope in the image. Recall that a single head of attention focuses on inter-patch relationships only, but two heads can potentially gauge how a composite of 2 or 3 patches relates to another composite of 2 or 3 patches. With multiple heads, the information captured tends to contain global context as opposed to neighboring or local relationships.

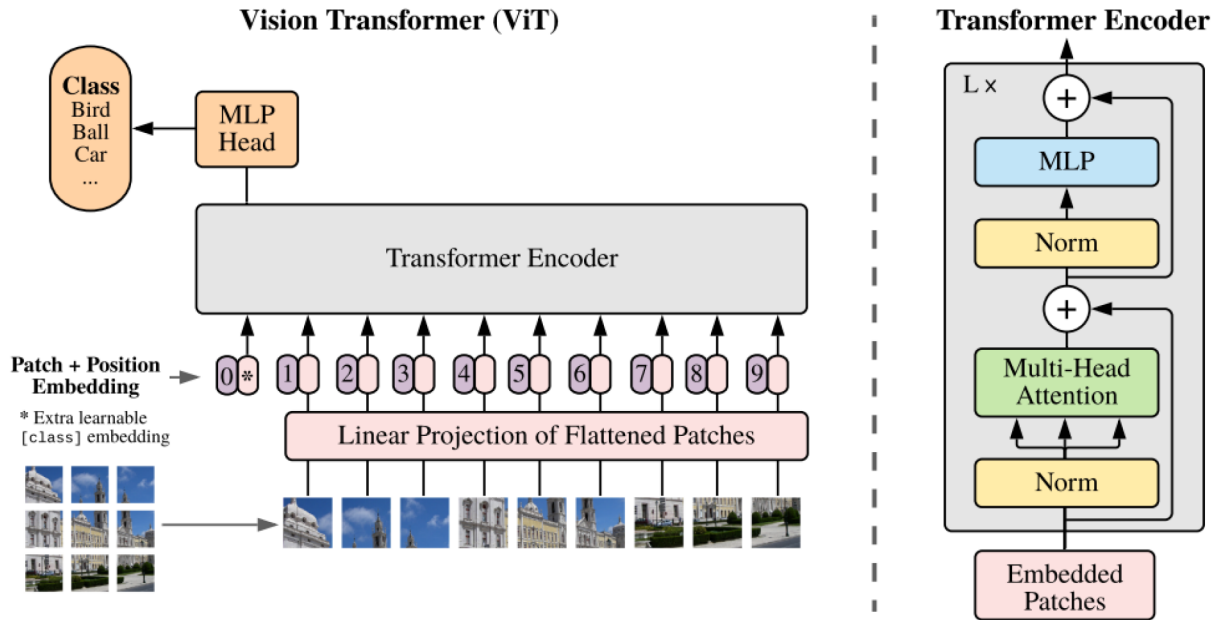


Figure 2—Vision Transformers architecture adapted from Dosovitskiy et al. (2020). In our work, we only use a Transformer Encoder.

A single ViT can be trained for classification tasks using labeled data (Bazi et al., 2021; Karuppasamy, 2023; Kim& Chung, 2024). As with any supervised learning task, this would involve successively feeding labeled images to the ViT and adjusting model parameters to minimize discrepancies between the model predictions and the labels. However, in this paper, we demonstrate how two ViT operating in parallel as teacher and student networks can automatically extract features by progressively learning from one another during training on unlabeled images, which is without supervision.

Feature Extraction with Self-Dstillation

Next, we demonstrate how a pair of initially untrained student and teacher ViTs operating in tandem allows for knowledge distillation from one network to the other and enables classification of images. The specific algorithm we use in this work is DINO (Caron et al., 2021). As shown in Fig. 3, DINO architecture consists of two ViT networks: the teacher and the student ViT network. Both networks are independently and randomly initialized at the start of the training process.

In Fig. 3, DINO processes an input image $\mathbf{x}$, generating two augmented views, $\mathbf{x1}$ and $\mathbf{x2}$. Image augmentation is a technique that introduces variation by modifying the appearance of an image to provide diverse learning perspectives (Goodfellow et al., 2016; Wang and Qi, 2022). To create these augmented views (image $\mathbf{x1}$ and image $\mathbf{x2}$) from the same image $\mathbf{x}$, DINO randomly applies augmentation such as horizontal flipping, brightness adjustment, and Gaussian blur, ensuring the student and teacher networks receive distinct yet related representations of the image. Additionally, DINO employs different cropping strategies: the student network receives a local crop (covering less than half of the augmented image), while the teacher network is given a global crop (covering more than half of the augmented image) (Caron et al., 2021).

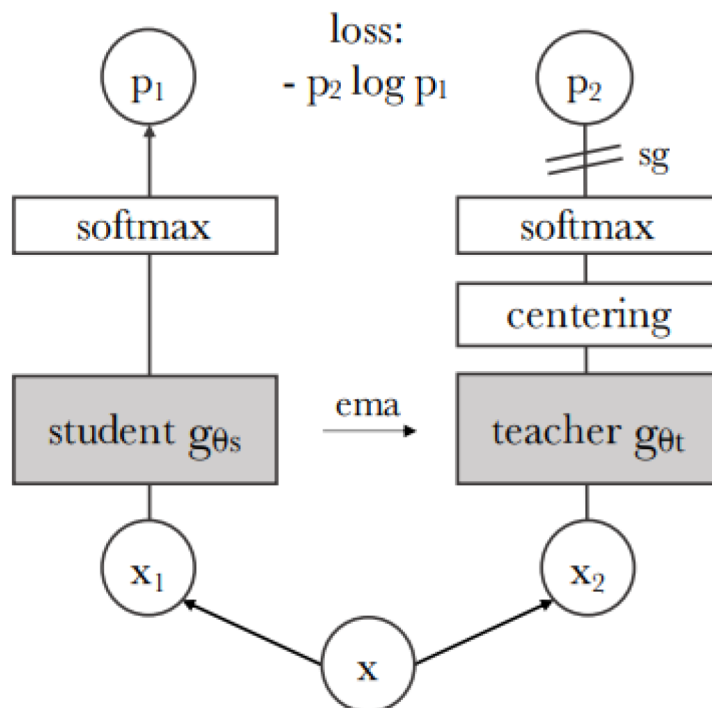


Figure 3—DINO architecture consists of two networks: a student and a teacher. Both are identical ViT networks but have different parameter sets (Caron et al., 2021)

Image x_1 and x_2 then pass through the student and teacher network, respectively, to produce the probability distributions, p_1 and p_2 , corresponding to the image classes. Prior to and during the training, p_1 and p_2 will likely be different, and the network uses these differences to fine-tune the student and teacher parameters using backpropagation. The teacher network is further updated as a weighted sum of its own parameters and the student network parameters. Subsequent training with additional images ensures that the student and teacher networks output a similar label for the same input image x . By using local as well as global crops of an image, the networks learn both local and larger-scale representations that can be used for classification. It is important to reiterate that the input image x is not labeled. Instead, the networks learn to assign the images to the same class or label. There are a few other considerations in the model architecture to ensure that the teacher and student do not converge on other trivial solutions and refer the reader to the original paper by Caron et al. (2021).

The Training Process.

Caron et al. (2021) trained a ViT classifier using the ImageNet dataset (Deng et al., 2009) and released a set of pretrained weights that can be used computer vision problems. This pre-trained ViT can then be leveraged for a variety of downstream tasks, such as image classification, segmentation, and object detection (Elharrouss et al., 2024).

Utilizing a pretrained backbone offers several advantages. There are significant savings in cost and computational effort by avoiding the need to train a machine learning model from scratch. However, a pre-trained model cannot be generalized easily to datasets significantly different from the original training set. In our initial experiments, we utilized the published pretrained weights as the backbone and obtained a Top 1 classification accuracy of less than 5%. We provide an explanation of Top 1 classification accuracy in a later section. Clearly, the pre-trained DINO backbone is unable to generalize to grayscale SEM images in unconventional shales.

Table 1—k-NN classification accuracy using DINO pretrained on ImageNet

k-NN	Top 1 (%)
5	1.36
10	1.92
50	3.74
100	4.39
200	4.55
500	4.55

To address this limitation, we retrained the DINO model from scratch using 80% of the FIB-SEM dataset. Because the original DINO was developed for RGB images, we follow the procedure documented in [Pfaendler et al. \(2023\)](#) for our grayscale dataset. We also retain the original hyperparameters from Caron et al. (2021), but modified the data augmentation workflow to include random horizontal flipping, cropping, and Gaussian blur as well as solarization and brightness adjustments.

The resulting DINO model, trained specifically on FIB-SEM images, now serves as the backbone for the post-processing image classification task. This version, referred to as the FIB-SEM-backbone, encodes the FIB-SEM images in the form of CLS token vectors.

Final Step for Image Classification using a K-Nearest Neighbors Classifier

The k-Nearest Neighbors (k-NN) algorithm predicts the label of a new data point by performing a local search neighbors and assigning the majority class label among its k nearest neighbors ([Cover and Hart, 1967](#)). In this study, we use the k-NN as a post-processing step for image classification. Following the training step, we now have CLS token vectors associated with each training image. To classify a test image, we first compute its CLS token using the trained DINO FIB-SEM-backbone and then measure the similarity between this token and the CLS tokens of its nearest neighbor images using the dot product (Caron et al., 2021; [Pfaendler et al., 2023](#)). We then rank the neighbors based on similarity and select the **top1** and **top2** most similar ones to assign class labels to the test images. Finally, we compute the confusion matrix between the predicted and known test labels.

For illustration, in a 5-nearest neighbors (5-NN) approach, the classification process for a test image x follows the steps outlined in [Table 2](#).

We test various values of k (5, 10, 50, 100, 200, and 500) and measure the **top1** and **top2** prediction accuracies for each case. The results, as presented in [Table 3](#), demonstrate that the model performance is consistent and remains highly accurate across all k values, with only minor variations. These results indicate that the model reliably predicts the correct outcome regardless of the number of nearest neighbors considered. Additionally, the consistently high **top2** accuracy reflects the ability of the model to predict the correct class among its two most probable predictions.

Table 2—Steps for predicting the label associated with an image ‘x’ using k-NN classifier

Steps	Example																								
Compute the classification token (feature representation) for image x .																									
Compute the dot product between image x 's classification token and all stored training tokens to measure similarity. Identify the top 5 highest similarity scores, representing the 5-nearest neighbors (5-NN) of image x .	<table><tr><th>Training Image</th><th>Class</th><th>Similarity Score</th></tr><tr><td>Image A</td><td>1</td><td>0.9</td></tr><tr><td>Image B</td><td>2</td><td>0.8</td></tr><tr><td>Image C</td><td>3</td><td>0.7</td></tr><tr><td>Image D</td><td>1</td><td>0.6</td></tr><tr><td>Image E</td><td>1</td><td>0.5</td></tr></table>	Training Image	Class	Similarity Score	Image A	1	0.9	Image B	2	0.8	Image C	3	0.7	Image D	1	0.6	Image E	1	0.5						
Training Image	Class	Similarity Score																							
Image A	1	0.9																							
Image B	2	0.8																							
Image C	3	0.7																							
Image D	1	0.6																							
Image E	1	0.5																							
Apply temperature scaling (T=0.1) to weight the top 5 nearest neighbor (5-NN) images based on their similarity scores.	<table><tr><th>Training Image</th><th>Similarity Score</th><th>Weight</th></tr><tr><td>Image A</td><td>0.9</td><td>$\exp(0.9/0.1) = 8103$</td></tr><tr><td>Image B</td><td>0.8</td><td>$\exp(0.8/0.1) = 2981$</td></tr><tr><td>Image C</td><td>0.7</td><td>$\exp(0.7/0.1) = 1096$</td></tr><tr><td>Image D</td><td>0.6</td><td>$\exp(0.6/0.1) = 403$</td></tr><tr><td>Image E</td><td>0.5</td><td>$\exp(0.5/0.1) = 148$</td></tr></table>	Training Image	Similarity Score	Weight	Image A	0.9	$\exp(0.9/0.1) = 8103$	Image B	0.8	$\exp(0.8/0.1) = 2981$	Image C	0.7	$\exp(0.7/0.1) = 1096$	Image D	0.6	$\exp(0.6/0.1) = 403$	Image E	0.5	$\exp(0.5/0.1) = 148$						
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Image D	0.6	$\exp(0.6/0.1) = 403$																							
Image E	0.5	$\exp(0.5/0.1) = 148$																							
Generate a one-hot encoding for the classes of the top 5 nearest neighbors (5-NN).	<table><tr><th>Class</th><th>Image A</th><th>Image B</th><th>Image C</th><th>Image D</th><th>Image E</th></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>2</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>3</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr></table>	Class	Image A	Image B	Image C	Image D	Image E	1	1	0	0	1	1	2	0	1	0	0	0	3	0	0	1	0	0
Class	Image A	Image B	Image C	Image D	Image E																				
1	1	0	0	1	1																				
2	0	1	0	0	0																				
3	0	0	1	0	0																				
Calculate the weighted score for each class based on the assigned weights.	<table><tr><th>Class</th><th>Computation of the Weighted Score</th><th>Weighted Score</th></tr><tr><td>1</td><td>$(8103 \times 1) + (0 \times 2981) + (0 \times 1096) + (403 \times 1) + (148 \times 1)$</td><td>8654</td></tr><tr><td>2</td><td>$(0 \times 8103) + (2981 \times 1) + (0 \times 1096) + (0 \times 403) + (0 \times 148)$</td><td>2981</td></tr><tr><td>3</td><td>$(0 \times 8103) + (0 \times 2981) + (1096 \times 1) + (0 \times 403) + (0 \times 148)$</td><td>1096</td></tr></table>	Class	Computation of the Weighted Score	Weighted Score	1	$(8103 \times 1) + (0 \times 2981) + (0 \times 1096) + (403 \times 1) + (148 \times 1)$	8654	2	$(0 \times 8103) + (2981 \times 1) + (0 \times 1096) + (0 \times 403) + (0 \times 148)$	2981	3	$(0 \times 8103) + (0 \times 2981) + (1096 \times 1) + (0 \times 403) + (0 \times 148)$	1096												
Class	Computation of the Weighted Score	Weighted Score																							
1	$(8103 \times 1) + (0 \times 2981) + (0 \times 1096) + (403 \times 1) + (148 \times 1)$	8654																							
2	$(0 \times 8103) + (2981 \times 1) + (0 \times 1096) + (0 \times 403) + (0 \times 148)$	2981																							
3	$(0 \times 8103) + (0 \times 2981) + (1096 \times 1) + (0 \times 403) + (0 \times 148)$	1096																							
Determine the predicted class for image x . In this case, image x is classified as Class 1, as it has the highest weighted score. In this example, the top 1 class is Class 1, while the top 2 classes consist of Class 1 and Class 2.	<table><tr><th>Class</th><th>Weighted Score</th></tr><tr><td>1</td><td>8654</td></tr><tr><td>2</td><td>2981</td></tr><tr><td>3</td><td>1096</td></tr></table>	Class	Weighted Score	1	8654	2	2981	3	1096																
Class	Weighted Score																								
1	8654																								
2	2981																								
3	1096																								

Table 3—k-NN classifier results for various k-values using FIB-SEM-backbone

k-NN	Top1 (%)	Top2 (%)
5	96	97
10	96	97
50	96	98
100	96	98
200	95	98
500	95	97

Fig. 4 shows the model classification performance using confusion matrices for 5-nearest neighbors. The confusion matrices for other choices of k for the kNN algorithm were similar and not shown here for brevity. In Fig. 4 the diagonal elements of the confusion matrix indicate the number of correct predictions, and off-diagonal terms represent misclassifications, which are relatively sparse.

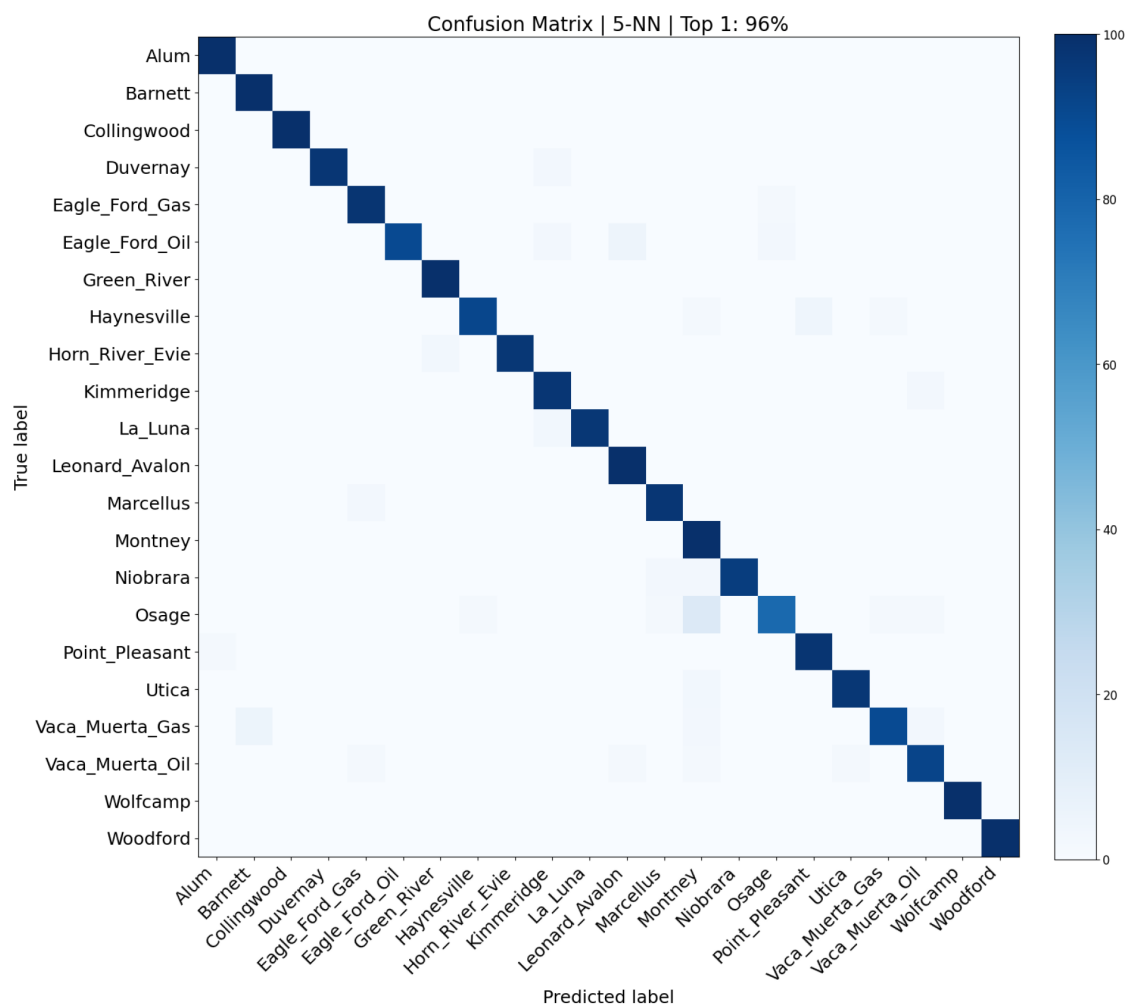


Figure 4—Confusion matrix for 5-NN classification model with an accuracy of 96%.

Model Interpretation

In traditional convolution neural network-based classifiers, deep learning features acquired from the last layer of the network contain globally relevant information of the input image. On the other hand, the intermediate feature maps point to local features necessary for classification (Prince, 2023). In vision transformers, this information is contained in an attention map that provides a visual representation of how the model interprets and prioritizes different regions of an image (Dosovitskiy et al., 2020; Caron et al., 2021). Our model consists of 12 transformer blocks stacked sequentially. Each block contains a multi-head self-attention mechanism that captures relationships between image patches, enabling the model to learn both local and global dependencies. As the input passes through each block, the model progressively constructs more abstract and comprehensive representations. For a given input image, we extracted the attention maps from the final transformer block to visualize the regions that the ViT considers most relevant for classification.

Fig. 5 presents an example of an attention map for a sample from the Collingwood play. The SEM image shows inorganic material in light gray at the top and some other inorganic material in a darker shade at

the bottom, separated by a fracture. Pores appear as dark spots in the bottom left. Notably, Heads 2 and 4 focus on the fracture, Head 6 highlights the pores, and Head 9 captures the inorganic material, clearly distinguishing it from the other inorganic regions. In Fig. 6, the SEM image from the Duvernay play depicts inorganic material in two different shades of gray, with organic matter embedded as darker gray areas within the lighter gray inorganic material. Here, Head 0 highlights the darker gray inorganic regions, Heads 1 and 9 capture the lighter gray areas, and Head 10 identifies the organic material embedded within the inorganic matrix.

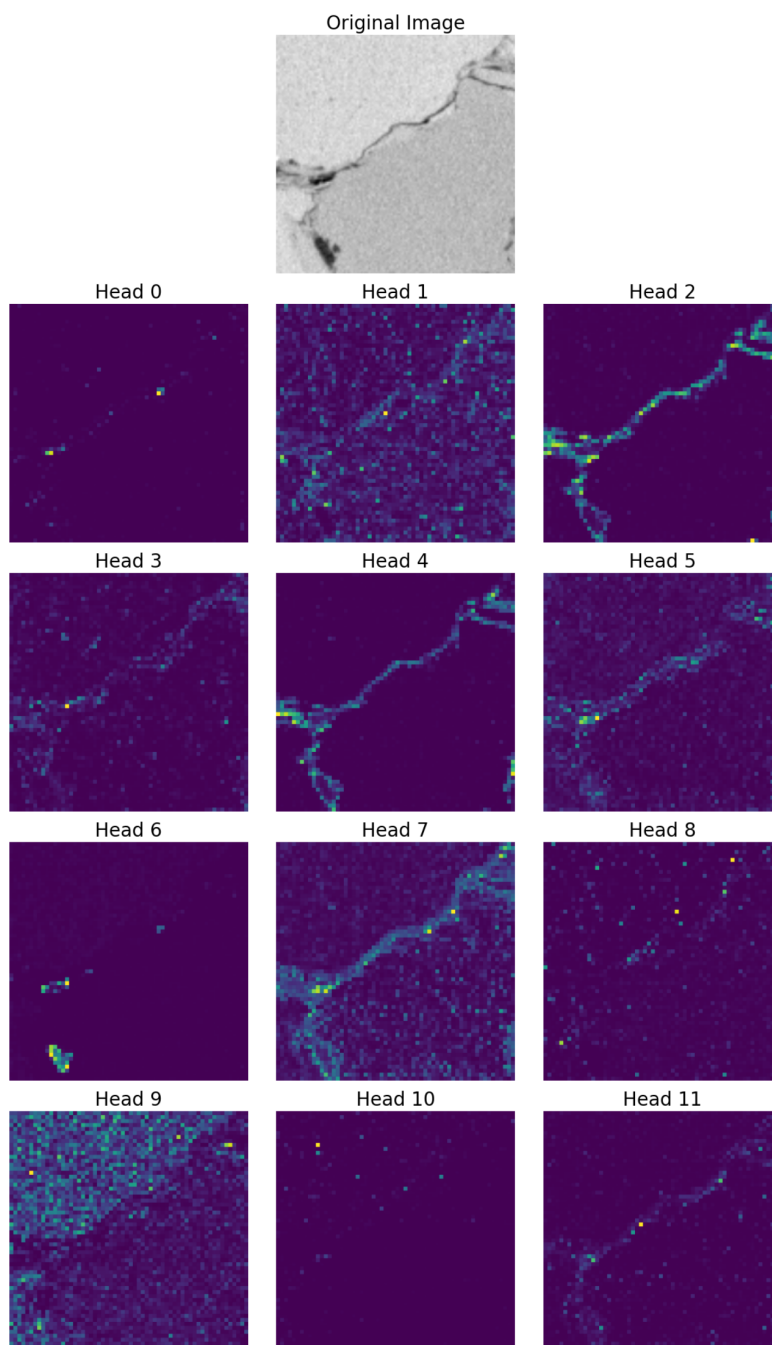


Figure 5—Example of attention map for a sample from Collingwood play. The numbers represent the output of each of the 12 attention heads in the DINO network. We can clearly see that the attention plots show that the networks quickly learn to identify key features (fracture in Head 2 and 4, pores in Head 6, inorganic in Head 9) without any training.

More notably, these attention maps were generated from test images without any prior training. They demonstrate how the model leverages interactions between patches, as outlined in the ViT framework, to recognize patterns and features within the images. These identified features are then aggregated into a classification token, which enables effective image classification. This visualization offers insight into the underlying model reasoning, showcasing its ability to differentiate between various textures and patterns, ultimately contributing to high classification accuracy.

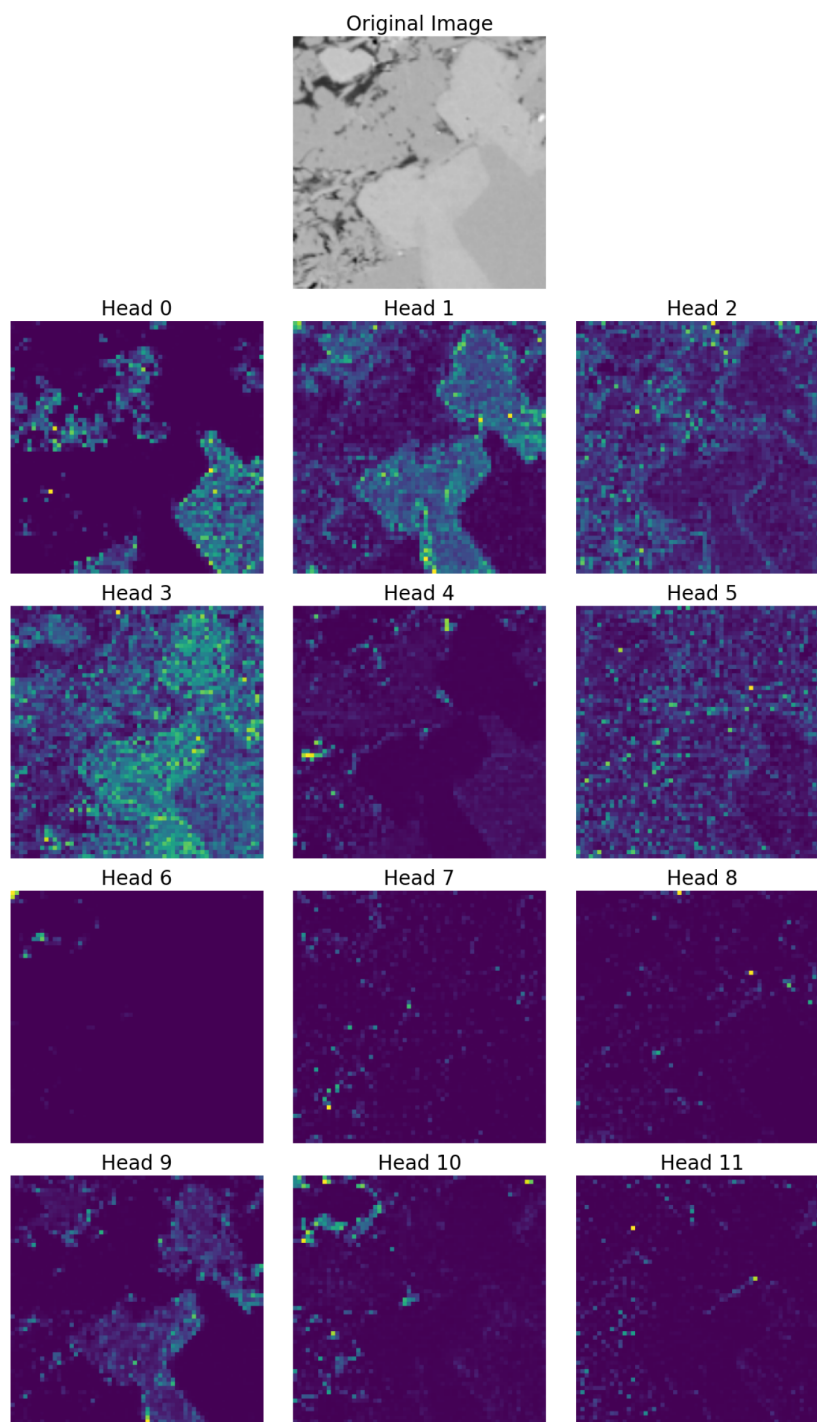


Figure 6—Example of attention map for a sample from Duvernay play for each of the 12 attention heads in the DINO network. Similar as Fig. 7, the networks quickly learn to identify key features (inorganic in Head 0 and 1, organic in Head 10), without any training.

Examples of Misclassified Images.

In this section, we discuss a few misclassified samples and attempt to showcase what might be causing the errors. While the misclassification rate is less than 5%, it might be instructive to understand the underlying cause. The first row in Fig. 7 displays images that were misclassified, while the second row shows randomly selected images associated with the predicted label for comparison. These comparisons showing similarities in grayscale patterns indicate why some images are misclassified.

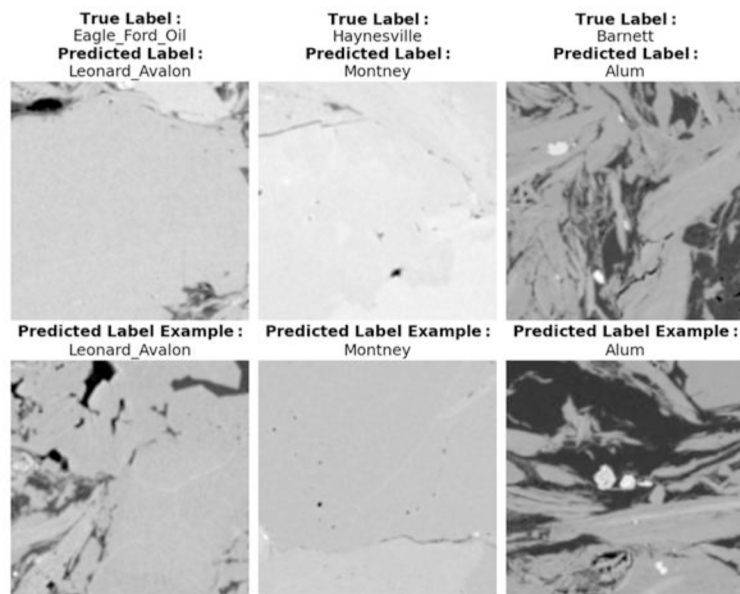


Figure 7—Examples of misclassified images. The first row displays the misclassified images along with their true labels and corresponding predicted labels. The second row provides reference examples of the predicted labels for comparison.

Conclusions

In this study, we explore the application of self-supervised learning with a model based on a vision transformer architecture for classifying shale SEM images and the following conclusions can be drawn from this work:

1. Self-supervised classification using a ViT architecture shows consistent and high accuracy when tested across 22 different shale plays.
2. The model consistently achieved high classification accuracy that is remarkable despite subtle variations in imaging conditions and microstructural similarities between formations.
3. The integration of attention mechanisms within the ViT model provides valuable insights into the features prioritized by the model during learning.

These findings demonstrate the potential of self-supervised ViT approaches in overcoming the limitations of traditional supervised methods, particularly the dependency on large human-annotated datasets. Moreover, this study highlights the value of combining self-supervised learning with interpretability tools, such as attention visualizations, to enhance our understanding of complex datasets like SEM images.

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